MINIMUM LAP TIME OPTIMIZATION

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Abstract.

Contents

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1. The problem setup

The objective of this project is to compute the minimum-time trajectory for a race car on a closed racing circuit. Unlike conventional path-following problems, where the trajectory is predefined, both the vehicle trajectory and the control inputs are optimized simultaneously while satisfying the physical limitations of the vehicle and the geometric constraints imposed by the track.

The optimization is performed on a real karting circuit located in Freiburg. The track geometry is represented by its centerline, local curvature and track width, which constitute the geometric input to the optimal control problem. These quantities are obtained by processing an image of the circuit and extracting its boundaries.

Besides the track geometry, the relevant vehicle parameters are assumed to be known. These include the wheelbase, steering angle limits, longitudinal acceleration and braking limits, and the maximum admissible lateral acceleration. Together, these parameters define the dynamic and physical constraints of the optimization problem.

The objective is therefore to determine the state and control trajectories that minimize the total lap time while ensuring that the vehicle remains within the track boundaries and satisfies all vehicle constraints.

To formulate this optimal control problem, an appropriate mathematical model of the vehicle is required. In the following section, a simplified kinematic bicycle model is introduced together with the assumptions adopted throughout this work.

2. Vehicle model

The vehicle model must capture the main relationship between steering, longitudinal motion, and the resulting trajectory while remaining sufficiently simple for the numerical optimization problem. Since this project focuses on the minimum-lap-time optimization problem rather than on detailed vehicle dynamics, we use a kinematic bicycle model.

The kinematic bicycle model describes the vehicle motion using only a few states and control inputs. It is simple enough for numerical optimization while still capturing the main effects of steering and acceleration. More detailed effects, such as tire slip, load transfer, suspension motion, and transient tire forces, are neglected.

2.1. The bicycle model.

The commonly used kinematic bicycle model is a simplification of the dynamics of a car. It represents the two wheels of each axle by a single equivalent wheel, resulting in one front wheel and one rear wheel. Only the front wheel is used for steering, and tire slip is neglected.

The model is described by the position of the car in global coordinates (x, y), the velocity v, the heading ψ, and the steering angle δ. The dynamics of the model are given by

$$\begin{aligned}
\dot{x} &= v\cos(\psi) \\
\dot{y} &= v\sin(\psi) \\
\dot{\psi} &= \frac{v}{L}\tan(\delta) \\
\dot{v} &= a
\end{aligned}$$

The parameter L denotes the wheelbase of the car, i.e. the distance between the front and rear wheel, and the acceleration of the car respectively. For more details on this model, see [Raj12].

However, for driving around a given track it is inconvenient to use global coordinates because it is more complicate to formulate the constraints. Additionally, we only need the position on the track and do not care for the absolute position in global coordinates.

Therefore, we want to formulate the model using track coordinates. The centerline distance traveled will be given as s and e_y denotes the distance from the centerline, e_ψ denotes the angle between the cars heading and the tangent of the centerline and v still denotes the velocity of the vehicle.

The position of the vehicle can be expressed relative to the track centerline as

$$\mathbf{p}(t) = \mathbf{p}_c(s(t)) + e_y(t)\mathbf{N}(s(t)),$$

where $\mathbf{p}_c(s)$ denotes the centerline position and $\mathbf{N}(s)$ is its unit normal vector.

Differentiating with respect to time gives

$$\dot{\mathbf{p}} = \frac{d\mathbf{p}_c}{ds}\dot{s} + \dot{e}_y\mathbf{N} + e_y\frac{d\mathbf{N}}{ds}\dot{s}.$$

Using

$$\frac{d\mathbf{p}_c}{ds} = \mathbf{T}, \qquad \frac{d\mathbf{N}}{ds} = -\kappa(s)\mathbf{T},$$

where $\mathbf{T}(s)$ is the unit tangent vector, we obtain

$$\dot{\mathbf{p}} = \left(1 - \kappa(s)e_y\right)\dot{s}\,\mathbf{T} + \dot{e}_y\,\mathbf{N}.$$

The vehicle velocity can also be decomposed into components tangent and normal to the centerline:

$$\dot{\mathbf{p}} = v\cos(e_\psi)\mathbf{T} + v\sin(e_\psi)\mathbf{N}.$$

Equating the tangential and normal components gives

$$\begin{aligned}\left(1 - \kappa(s)e_y\right)\dot{s} &= v\cos(e_\psi),\\ \dot{e}_y &= v\sin(e_\psi).\end{aligned}$$

Therefore,

$$\dot{s} = \frac{v\cos(e_\psi)}{1 - \kappa(s)e_y}.$$

This relation is commonly used in Frenet-coordinate formulations of vehicle dynamics [RNFD23].

The heading error is defined as

$$e_\psi = \psi - \psi_c(s),$$

where $\psi_c(s)$ denotes the heading of the track centerline.

Differentiating with respect to time and applying the chain rule yields

$$\dot{e}_\psi = \dot{\psi} - \frac{d\psi_c}{ds}\dot{s}.$$

The cars own heading with respect to the tangent of the centerline changes because of steering and the heading of the track changes because of the curvature of the track.

Since the derivative of the centerline heading with respect to the arc length is equal to the local track curvature,

$$\frac{d\psi_c}{ds} = \kappa(s),$$

the heading-error dynamics become

$$\dot{e}_\psi = \frac{v}{L}\tan(\delta) - \kappa(s)\dot{s}.$$

This results in the following vehicle model expressed in Frenet coordinates. The state vector is defined as

$$z(t) = \begin{bmatrix} s(t) \\ e_y(t) \\ e_\psi(t) \\ v(t) \end{bmatrix},$$

where $s(t)$ denotes the progress along the track centerline, $e_y(t)$ the lateral offset from the centerline, $e_\psi(t)$ the heading error, and $v(t)$ the vehicle velocity.

The control vector is given by

$$u(t) = \begin{bmatrix} a(t) \\ \delta(t) \end{bmatrix},$$

where $a(t)$ is the longitudinal acceleration and $\delta(t)$ is the steering angle.

The state dynamics can then be written compactly as

$$\dot{z}(t) = f\big(z(t), u(t)\big) = \begin{bmatrix} \dfrac{v(t)\cos(e_\psi(t))}{1 - \kappa(s(t))e_y(t)} \\ v(t)\sin(e_\psi(t)) \\ \dfrac{v(t)}{L}\tan(\delta(t)) - \kappa(s(t))\dfrac{v(t)\cos(e_\psi(t))}{1 - \kappa(s(t))e_y(t)} \\ a(t) \end{bmatrix}.$$

The kinematic bicycle model neglects tire slip, load transfer, suspension dynamics and transient tire forces. Consequently, it is not intended to accurately describe the vehicle behaviour at the limits of adhesion.

Nevertheless, it captures the main relationship between steering, longitudinal acceleration and the resulting vehicle trajectory while keeping the optimization problem sufficiently simple.

3. Optimization Problem

3.1. **Formulation of the optimization problem.**

Using the vehicle model introduced in the previous section, the minimum-lap-time problem can be formulated as a continuous-time optimal control problem. The decision variables are the state trajectory $z(\cdot)$, the control trajectory $u(\cdot)$, and the final time T.

The control vector

$$u(t) = \begin{bmatrix} a(t) \\ \delta(t) \end{bmatrix}$$

contains the longitudinal acceleration and the steering angle.

The objective is to minimize the total time required to complete the track. The resulting optimal control problem is

$$\underset{z(\cdot),u(\cdot),T}{\text{minimize}} \quad T \tag{3.1}$$

$$\text{subject to} \quad \dot{z}(t) = f\big(z(t), u(t)\big), \qquad t \in [0,T], \tag{3.2}$$

$$-\frac{w\big(s(t)\big)-b}{2} \le e_y(t) \le \frac{w\big(s(t)\big)-b}{2}, \qquad t \in [0,T], \tag{3.3}$$

$$-a_{\text{min}} \le a(t) \le a_{\text{max}}, \qquad t \in [0,T], \tag{3.4}$$

$$-\delta_{\text{max}} \le \delta(t) \le \delta_{\text{max}}, \qquad t \in [0,T], \tag{3.5}$$

$$0 \le v(t) \le v_{\text{max}}, \qquad t \in [0,T], \tag{3.6}$$

$$-a_{y,\text{max}} \le \frac{v(t)^2}{L}\tan\big(\delta(t)\big) \le a_{y,\text{max}}, \qquad t \in [0,T], \tag{3.7}$$

$$1-\kappa\big(s(t)\big)e_y(t) > 0, \qquad t \in [0,T], \tag{3.8}$$

$$s(0)=0, \qquad s(T)=S, \tag{3.9}$$

$$v(0)=0, \qquad e_y(0)=0, \qquad e_\psi(0)=0, \tag{3.10}$$

$$T > 0. \tag{3.11}$$

The parameters appearing in the optimal control problem are defined as follows:

$w(s)$: track width at centerline position s,

b: vehicle width,

L: vehicle wheelbase,

$a_{\text{min}} > 0$: magnitude of the maximum permissible deceleration,

$a_{\text{max}} > 0$: maximum permissible acceleration,

δ_{max}: maximum absolute steering angle,

v_{max}: maximum permissible velocity,

$a_{y,\text{max}}$: maximum permissible lateral acceleration,

S: total centerline length of the track.

The objective function (3.1) minimizes the final time T. The dynamic constraint (3.2) ensures that the state trajectory is consistent with the kinematic bicycle model.

The track constraint (3.3) restricts the center of the vehicle to the usable track width. The vehicle width b is subtracted from the track width so that the complete vehicle, rather than only its reference point, remains inside the track boundaries.

The acceleration, steering, and velocity constraints (3.4)–(3.6) represent the longitudinal and kinematic limitations of the vehicle. The lateral acceleration constraint (3.7) limits the cornering acceleration generated by the steering input.

The boundary conditions require the vehicle to start at the beginning of the track and reach the total centerline distance S. In the present formulation, the vehicle starts from rest at the centerline with zero heading error.

3.2. **Extension: Static obstacle avoidance.**

So far, the optimization problem only considers the track boundaries and vehicle limitations. In practical racing scenarios, however, additional objects may occupy parts of the track. To account for such situations, the formulation can be extended by introducing static obstacles represented in Frenet coordinates.

The positions of the obstacles are assumed to be known in advance and are specified in Frenet coordinates. For each obstacle $i = 1, \dots, N_{\text{obs}}$, the centerline position is denoted by s_i and the lateral offset by $e_{y,i}$. Furthermore, each obstacle is assigned a length l_i and a width w_i.

To account for the finite dimensions of both the obstacle and the vehicle, together with an additional safety margin, each obstacle is surrounded by an elliptical safety region. The corresponding semi-axes in the longitudinal and lateral directions are defined as

$$r_{s,i} = \frac{l_i}{2} + \frac{L}{2} + d_{\text{safe}},$$
$$r_{e_y,i} = \frac{w_i}{2} + \frac{b}{2} + d_{\text{safe}},$$

where L denotes the vehicle wheelbase, b its width, and d_{safe} the prescribed safety margin.

For every obstacle, the vehicle is required to remain outside the corresponding safety region. This requirement is enforced by introducing the nonlinear constraint

$$\left(\frac{s - s_i}{r_{s,i}}\right)^2 + \left(\frac{e_y - e_{y,i}}{r_{e_y,i}}\right)^2 \geq 1, \qquad i = 1, \dots, N_{\text{obs}}. \tag{3.12}$$

Since the optimization problem is solved on a discrete time grid, obstacle avoidance constraints are additionally evaluated at the midpoint of every integration interval. This improves collision detection between consecutive discretization nodes while avoiding the computational expense of using a much finer discretization.

This extension illustrates how additional practical constraints can be integrated into the optimal control formulation without modifying the underlying vehicle model.

4. Numerical solution

The optimal control problem is solved using a direct multiple shooting approach [BP84]. The total lap time T is considered as an optimization variable and the time horizon is divided into $N = 200$ intervals of equal duration,

$$\Delta t = \frac{T}{N}. \tag{4.1}$$

The state vector

$$Z_k = [s_k, e_{y,k}, e_{\psi,k}, v_k]^{\mathrm{T}}$$

is defined at each of the $N + 1$ shooting nodes, while the control vector

$$U_k = [a_k, \delta_k]^{\mathrm{T}}$$

is assumed to remain constant over each interval.

To propagate the vehicle dynamics between two consecutive shooting nodes, a classical fourth-order Runge–Kutta (RK4) integration scheme is used. For each interval,

$$K_1 = f(Z_k, U_k), \tag{4.2}$$

$$K_2 = f\left(Z_k + \frac{\Delta t}{2} K_1, U_k\right), \tag{4.3}$$

$$K_3 = f\left(Z_k + \frac{\Delta t}{2} K_2, U_k\right), \tag{4.4}$$

$$K_4 = f\left(Z_k + \Delta t K_3, U_k\right), \tag{4.5}$$

and the state at the next node is approximated by

$$\widehat{Z}_{k+1} = Z_k + \frac{\Delta t}{6}\left(K_1 + 2K_2 + 2K_3 + K_4\right). \tag{4.6}$$

The continuity of the trajectory is enforced by requiring the optimization variables to satisfy

$$Z_{k+1} = \widehat{Z}_{k+1}, \qquad k = 0, \dots, N-1. \tag{4.7}$$

The optimization problem is implemented in Python using the CasADi framework and its `Opti` interface [AGHRD19]. CasADi provides symbolic modelling, automatic differentiation and a convenient interface to nonlinear optimization solvers. The track curvature is evaluated through a one-dimensional linear interpolant so that $\kappa(s)$ can be computed for any optimized position along the track.

The resulting nonlinear programming problem is solved with IPOPT [WB06]. In the present implementation, the maximum number of iterations is set to 3000, with a convergence tolerance of 10^{-7} and an acceptable tolerance of 10^{-5}.

As nonlinear optimization is sensitive to the initial guess, a simple physically meaningful initialization is provided. The vehicle is assumed to accelerate from rest to a constant cruising speed, while the initial steering angle is estimated from the centerline curvature as

$$\delta_k^{(0)} = \arctan\left(L\kappa(s_k^{(0)})\right). \tag{4.8}$$

The initial lateral displacement and heading error are both set to zero, meaning that the initial trajectory follows the track centerline.

5. Results

5.1. Original formulation.

The optimization problem was solved using $N = 200$ shooting intervals. IPOPT converged successfully after 357 iterations, requiring approximately 8.0 s of computation time. The solver returned the status `Solve_Succeeded`. The optimal lap time obtained was

$$T^* = 61.665 \text{ s}. \tag{5.1}$$

The main numerical results are summarized in Table 1.

TABLE 1. Main optimization results for $N = 200$.

Quantity	Result
Optimal lap time	61.665 s
Mean velocity	11.864 m/s
Maximum velocity	12.500 m/s
Lateral-offset range	$[-2.870,\ 2.870]$ m
Acceleration range	$[-6.000,\ 3.000]$ m/s^2
Steering-angle range	$[-0.600,\ 0.126]$ rad
Lateral-acceleration range	$[-8.000,\ 8.000]$ m/s^2
Minimum Frenet denominator	0.6827
IPOPT iterations	357
Computation time	7.996 s
Solver status	`Solve_Succeeded`

Figure 1 shows the reconstructed track together with the optimized trajectory. Rather than following the centerline, the vehicle moves across the available track width to obtain smoother cornering lines and reduce the lap time.

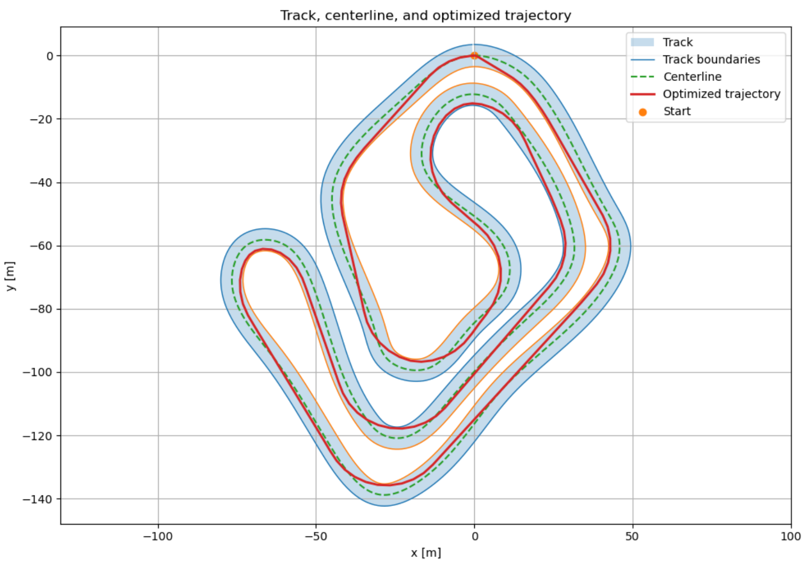

FIGURE 1. Reconstructed track geometry, centerline and optimized trajectory.

The corresponding lateral displacement is shown in Figure 2. The optimizer makes use of almost the entire admissible width, reaching the limits of ±2.87 m in several corners. This behaviour is consistent with an optimal racing line, where the vehicle moves towards the outside before entering a corner and approaches the inside near the apex.

The optimal velocity profile is presented in Figure 3. The kart rapidly accelerates to the maximum allowed speed of 12.5 m/s and remains close to this value over most of the circuit. Speed reductions only occur in the sections with the highest curvature, where the lateral-acceleration constraint becomes active.

The longitudinal acceleration varies between -6.0 and 3.0 m/s^2, while the lateral acceleration reaches the imposed limits of ± 8.0 m/s^2. This indicates that both the longitudinal

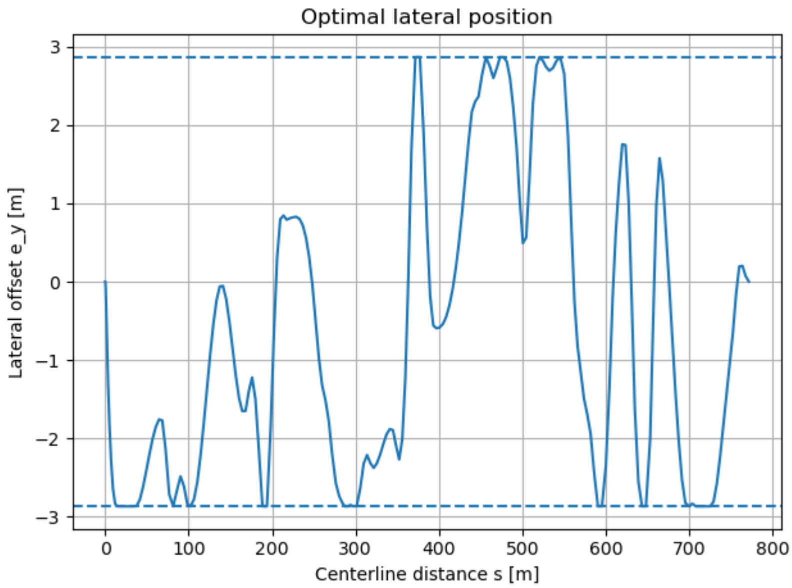

FIGURE 2. Optimized lateral displacement with admissible track limits.

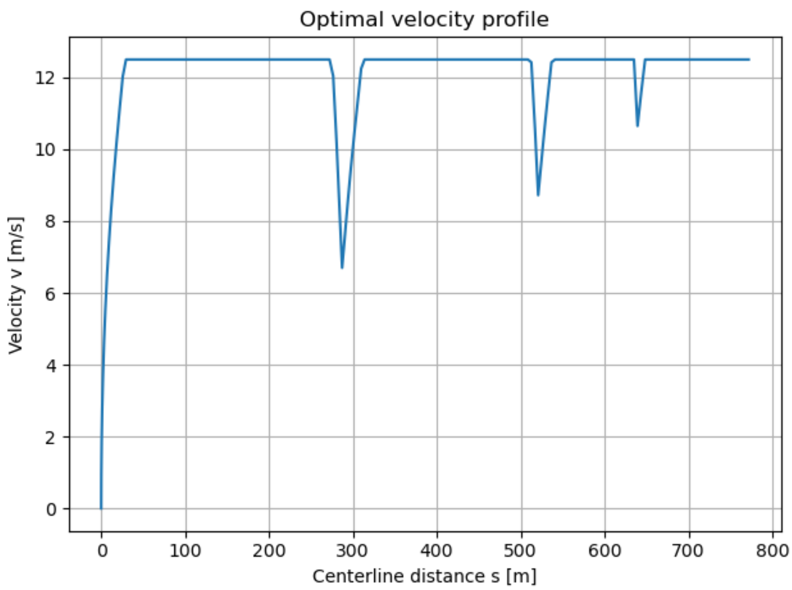

FIGURE 3. Optimal velocity profile along the track centerline.

and lateral capabilities of the vehicle are fully exploited during the lap. The steering angle remains within the prescribed bounds, ranging from -0.600 to 0.126 rad.

Finally, the minimum value of the Frenet-coordinate denominator is

$$\min(1 - \kappa e_y) = 0.6827,$$

which remains well above zero throughout the optimization. Therefore, the computed trajectory stays within the valid region of the Frenet-coordinate formulation.

5.2. **Extended formulation.**
For the extended formulation, seven static obstacles were placed along the track. Their

positions were chosen on the centerline at locations where the optimal trajectory of the original formulation crosses the track centerline, requiring the optimizer to generate meaningful avoidance maneuvers. The vehicle and track parameters remained identical to those used in the original formulation. Each obstacle was assigned a length of 1.5 m and a width of 0.8 m, together with a safety margin of 0.2 m. To improve the solution quality in the presence of the additional constraints, the number of discretization intervals was increased to $N = 300$.

Figure 4 shows the optimized trajectory together with the obstacle safety regions. The vehicle successfully avoids all seven obstacles while remaining within the track boundaries. A detailed view of several obstacle locations is presented in Figure 5, where the local trajectory modifications required for obstacle avoidance can be observed more clearly.

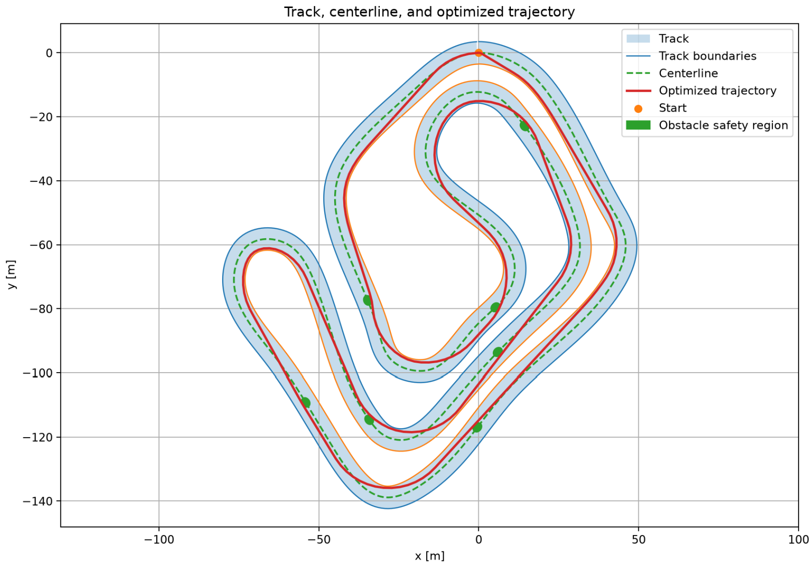

FIGURE 4. Track geometry with static obstacles.

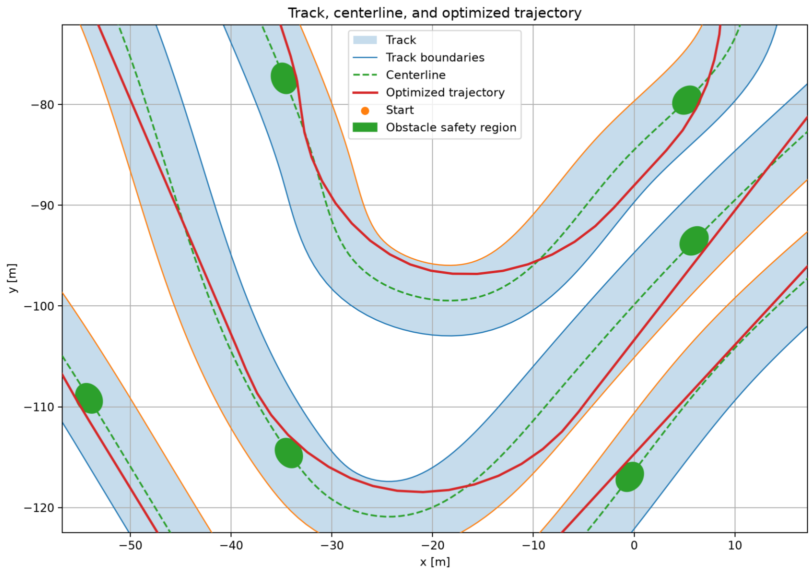

FIGURE 5. Close-up of the optimized trajectory near several obstacles.

The comparison of the lateral position profiles in Figure 6 shows that the additional constraints affect the trajectory only locally. Away from the obstacles, the solution closely follows that of the original formulation, while noticeable changes in the lateral position occur only in the vicinity of the obstacle locations.

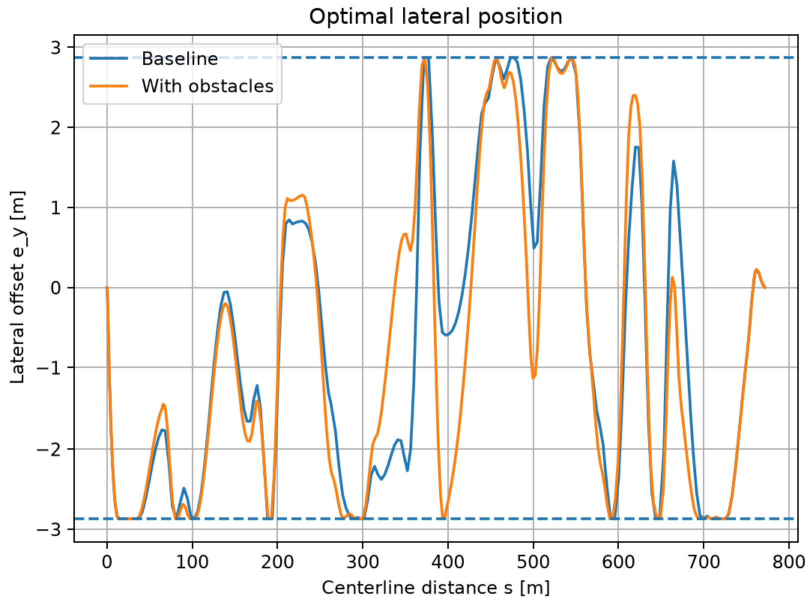

FIGURE 6. Comparison of optimal lateral position.

Overall, the optimized solution remains very similar to that of the original formulation. The optimal lap time increases only slightly, from 61.665 s to 61.800 s, while both the maximum and average velocities remain nearly unchanged. This indicates that the additional obstacle avoidance constraints have only a minor impact on the overall performance.

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